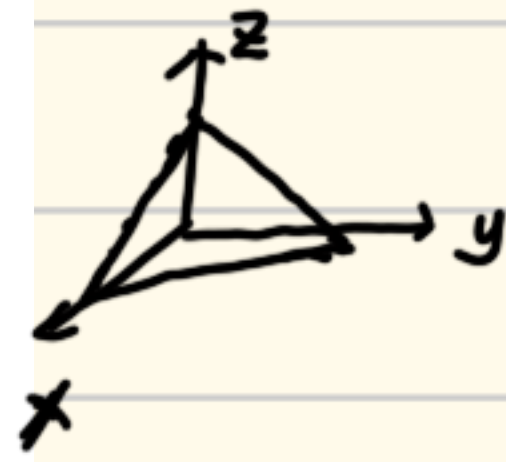


求下列第二型曲面积分：

1. 求 $\iint_S yz dy dz + xz dz dx + xy dx dy$ 其中 S 为平面 $x = 0, y = 0, z = 0$ 及 $x + y + z = a (a > 0)$ 所围四面体的表面外侧。
2. 求 $\iint_S xz dy dz + xy dz dx + yz dx dy$, 其中 S 是圆柱面 $x^2 + y^2 = R^2$ 与平面 $x = 0, y = 0, z = 0$ 及 $z = h (h > 0)$ 所围的在第一卦限中的立体的表面外侧。
3. 求 $\iint_S x^2 y^2 z dx dy$, S 是球面 $x^2 + y^2 + z^2 = R^2$ 下半部的下侧。
4. 求 $\iint_S z^2 dy dz + x dz dx - 3z dx dy$, 其中 S 为由曲面 $z = 4 - y^2$, 平面 $x = 0, x =$



xOz 面上: $y=0$

$$\begin{aligned} \iint_S xz dz dx &= \int_0^a dx \int_0^{a-x} xz dz \\ &= \frac{1}{2} \int_0^a (a-x)^2 x \cdot dx \\ &= \frac{1}{2} \int_0^a x^3 - 2ax^2 + a^2x dx \\ &= \frac{1}{2} \left(\frac{a^4}{4} - \frac{2a^4}{3} + \frac{1}{2}a^4 \right) \\ &= -\frac{1}{24}a^4, \text{ 同理在 } xOy, yOz \text{ 面上也轮换对称;} \end{aligned}$$

2. $z=0, x^2 + y^2 = R^2, z=h$

$$\begin{aligned} z=h, x^2+y^2=R^2, \iint_S yz dx dy \\ = h \int_0^{2\pi} d\theta \int_0^R r^2 \sin\theta \cdot dr \\ = \frac{1}{3}R^3h \end{aligned}$$

$x^2+y^2=R^2, 0 \leq z \leq h$

$$\iint_S xz dy dz + xy dz dx$$

$z = a - x - y$

$$\begin{aligned} &\iint_D [(x+y)^2 - a(x+y)] d\sigma \\ &= \int_0^a dx \int_0^{a-x} (x^2 + 2xy + y^2 - ax - ay) \cdot dy \\ &= \int_0^a \left[(a-x) \cdot x^2 + x(a-x)^2 + \frac{1}{3}(a-x)^3 - ax(a-x) - \frac{1}{2}a(a-x)^2 \right] dx \\ &= \int_0^a \left(\cancel{ax^2} - \cancel{x^3} + \cancel{x^3} - \cancel{2ax^2} + \boxed{a^2x} - \frac{1}{3}x^3 + \boxed{ax^2} - \boxed{a^2x} + \frac{1}{3}a^3 - \cancel{a^2x} + \boxed{ax^2} - \frac{1}{2}a^3 + \boxed{a^2x} - \boxed{\frac{ax^2}{2}} \right) dx \\ &= \int_0^a \left(-\frac{1}{3}x^3 - a^2x + \frac{3}{2}ax^2 - \frac{1}{6}a^3 \right) dx \\ &= 0 - \frac{a^4}{12} - \frac{1}{3}a^4 + \frac{1}{2}a^4 - \frac{1}{6}a^4 \\ &= -\frac{1}{12}a^4 \end{aligned}$$

$\frac{1}{2}ax^2 - \frac{1}{6}a^3 - \frac{1}{3}x^3$
 $= -\frac{1}{12}a^4 - \frac{1}{6}a^4 + \frac{1}{6}a^4$